PROJECT BOOKLET

HR Employee Attrition & Retention Dashboard

A Power BI analysis of why employees leave and who is most at risk

TEAM

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Tools: Power Query · DAX · Power BI Desktop | Version 1.0 – October 2026

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1,470	237	16.12%	83.9%
Total employees	Employees who left	Attrition rate	Retention rate

This booklet documents the full analysis cycle of the HR attrition project: from understanding and preparing the data, to modeling, KPIs, dashboard design, findings and recommendations.

1. Project Overview

This project analyzes employee attrition for a company of **1,470 employees**, of whom **237 left**, an attrition rate of **16.12%**. We built an interactive Power BI dashboard that lets HR explore which groups leave the most and what factors are linked to leaving, then turned the results into concrete recommendations.

Objectives

- Clean and prepare the HR employee data.
- Build a data model and the KPIs needed to measure attrition.
- Design an interactive dashboard with slicers, filters and page navigation.
- Turn the results into clear business insights and recommended actions.

Tools used

Stage	Tool
Data cleaning & preparation	Power Query (Power BI)
Data modeling & measures	Power BI, DAX
Dashboard design	Power BI Desktop
Documentation & presentation	PDF, PowerPoint
Version control	Git & GitHub

Team

Member	Role
Sandra Hany	Person 1: Data understanding & cleaning (Power Query, data dictionary)
Ahmed Samy	Person 2: Data modeling & DAX (model, relationships, KPIs)
Haneen Tarek	Person 3: Visualization & dashboard (pages, navigation, UX)
Arsany Hany	Person 4: Advanced analysis, validation & final dashboard
Alzahraa Hassan	Person 5: Final integration, documentation & delivery

2. Business Problem

Losing employees is costly: recruiting, training and the loss of knowledge all add up. With 237 of 1,470 employees gone, HR needs evidence about where to act first.

Questions the analysis answers

- Which employee groups leave the most?
- What drives attrition: overtime, pay, tenure, business travel, age, marital status?
- Which departments and job roles are most affected?
- What actions can reduce attrition?

3. Dataset Overview

Item	Detail
File	HR_Employee_Attrition_Cleaned
Rows	1,470 employees (one row per employee)
Employees who left	237 (16.12%)
Source	IBM HR Analytics Employee Attrition & Performance (Kaggle): kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset
Number of columns	32 (all snake_case names)
Data quality	0 missing values, 0 duplicate rows

Key fields

Group	Fields
Demographics	Age group, gender, marital status, education field
Job	Department, job role, job level
Compensation	Monthly income, income band, stock option level
Work conditions	Overtime, business travel, job satisfaction
Tenure & career	Years at company, years since last promotion, number of companies worked
Outcome	Attrition status (Left / Stayed)

The dataset is a public, fictional sample created by IBM data scientists, so it does not describe a real company.

4. Data Quality & Cleaning

The dashboard uses the cleaned file *HR_Employee_Attrition_Cleaned*. Cleaning was done in Power Query. The following preparation is visible in the final dashboard:

- Quality check on the cleaned file: **32 columns, 0 missing values, 0 duplicate rows**; column names are in snake_case (e.g. job_level, monthly_income).
- Attrition status converted to a readable **Left / Stayed** outcome.
- Analysis bands created from numeric columns: **age group** (Under 30, 30-39, 40-49, 50+), **income band** (Under 3K, 3K-6K, 6K-10K, 10K+), **tenure group** (0-1, 2-5, 6-10, 10+ years) and **years since last promotion** (0, 1, 2, 3, 4-5, 6+).

- Job level and business travel values labeled for readability (e.g. Travel Frequently / Travel Rarely / Non-Travel; Job Satisfaction 1 – Low to 4 – Very High).
- The public original has 35 columns; the cleaned file has 32. The three constant columns *EmployeeCount*, *Over18* and *StandardHours* are not in it, and business_travel values use spaces (Travel Rarely).

TODO – team input needed: Sandra (Person 1): confirm the three points above and send a screenshot of Power Query → Applied Steps, so the exact steps can be listed.

5. Data Modeling

Table	Purpose
Fact_Employee	One row per employee: demographics, income, tenure, attrition status
Dim_Job	Department, job role, job level
Measures_HR	Holds all DAX measures
Progress Axis	Helper table used for visuals in the dashboard

The model follows a simple star-style design: **Fact_Employee** is the central fact table and **Dim_Job** describes the job attributes. Measures live in a dedicated table so they are easy to find and reuse.

TODO – team input needed: Ahmed (Person 2): add the relationship details (keys, cardinality, filter direction) and the Data Model screenshot from Model view.

6. DAX & KPIs

Main KPIs shown across the dashboard pages:

KPI	Definition	Value
Total Employees	Count of employees	1,470
Active Employees	Employees who stayed	1,233
Attrition Rate	Employees who left ÷ total employees	16.12%
Retention Rate	Active employees ÷ total employees	83.9%
Average Monthly Income	Average monthly income of employees	$6,503
Avg Income – Attrited	Average monthly income of employees who left	$4,787
Avg Years at Company	Average tenure	7.01
Avg Years Since Promotion	Average years since last promotion	2.2
Avg Age	Average age in years	36.9
Overtime Attrition Rate	Attrition rate among employees working overtime	30.5%

Sample measures (illustrative)

```
Total Employees = COUNTROWS ( Fact_Employee )
Employees Left = CALCULATE ( [Total Employees], Fact_Employee[Attrition] = "Yes" )
Attrition Rate = DIVIDE ( [Employees Left], [Total Employees] )
Active Employees = [Total Employees] - [Employees Left]
```

TODO – team input needed: Ahmed (Person 2): replace the sample DAX above with the real measures from the .pbix (Measures_HR) and add the explanation of each. Column names above are placeholders.

7. Dashboard Overview

The report has seven pages. A left-hand navigation panel moves between pages, and each page has filter/clear-filter icons and slicers for exploration.

Page	What it shows
Home	Cover page and navigation
Overview	Headline KPIs, attrition by job level, department and age group, left vs stayed
Drivers	Attrition by tenure, income band, business travel, marital status and overtime
Compensation	Income analysis, attrition by job role, stock options and job satisfaction
Tenure	Years at company, promotions and number of companies worked
Workforce	Headcount by job role, age, education field, marital status and gender
Insights	Eight key findings, each with a recommended action



Figure 1 – Home page: cover and navigation.

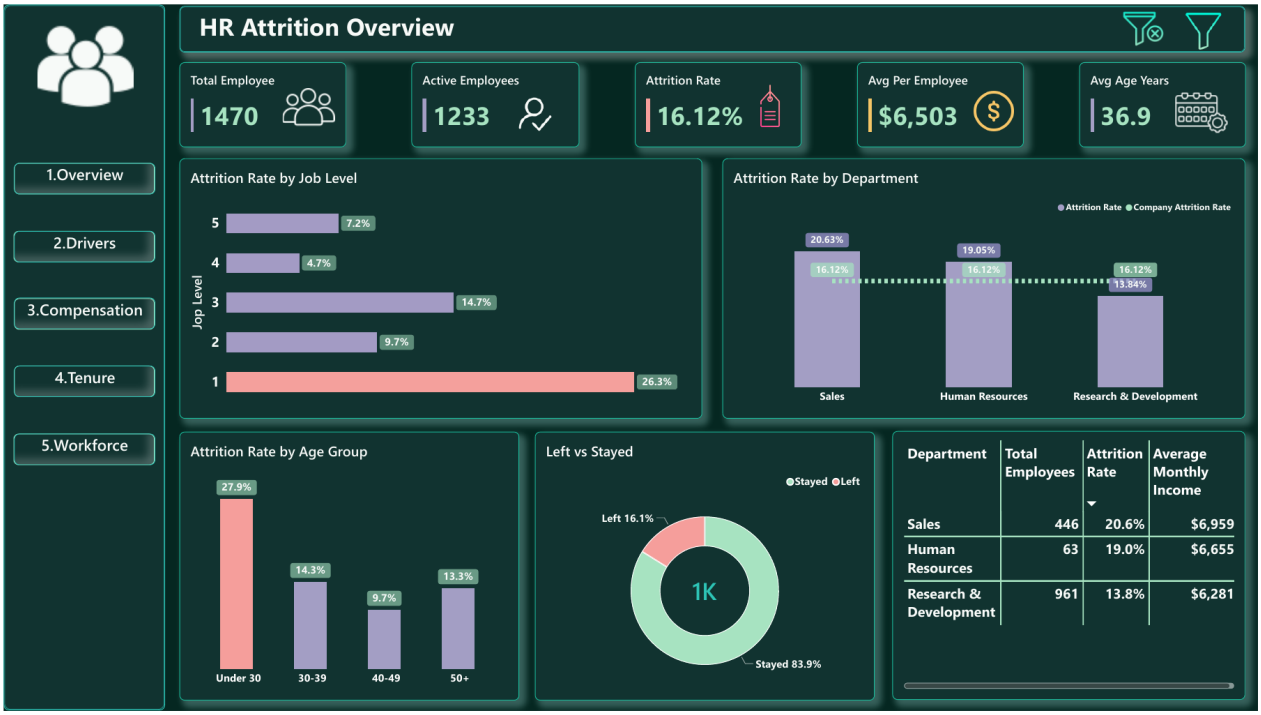


Figure 2 – Overview: headline KPIs and attrition by job level, department and age group.

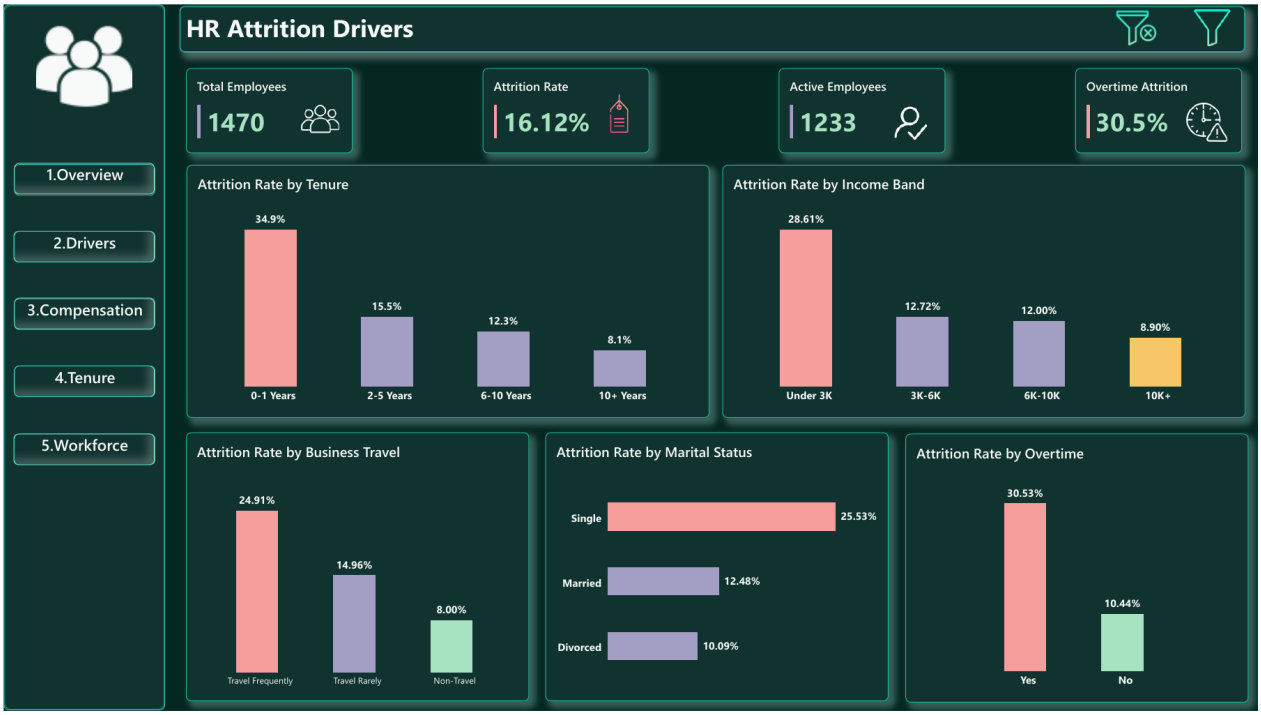


Figure 3 – Drivers: tenure, income, travel, marital status and overtime.

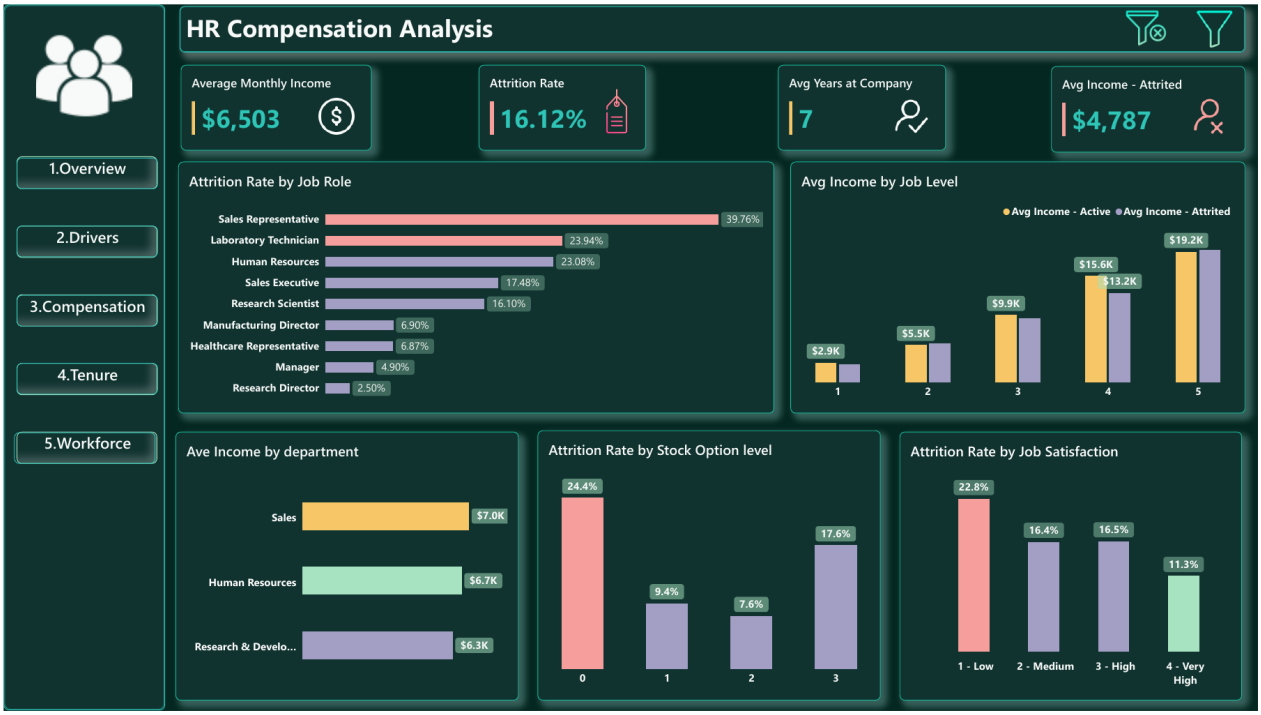


Figure 4 – Compensation: job role, income by level, stock options and job satisfaction.

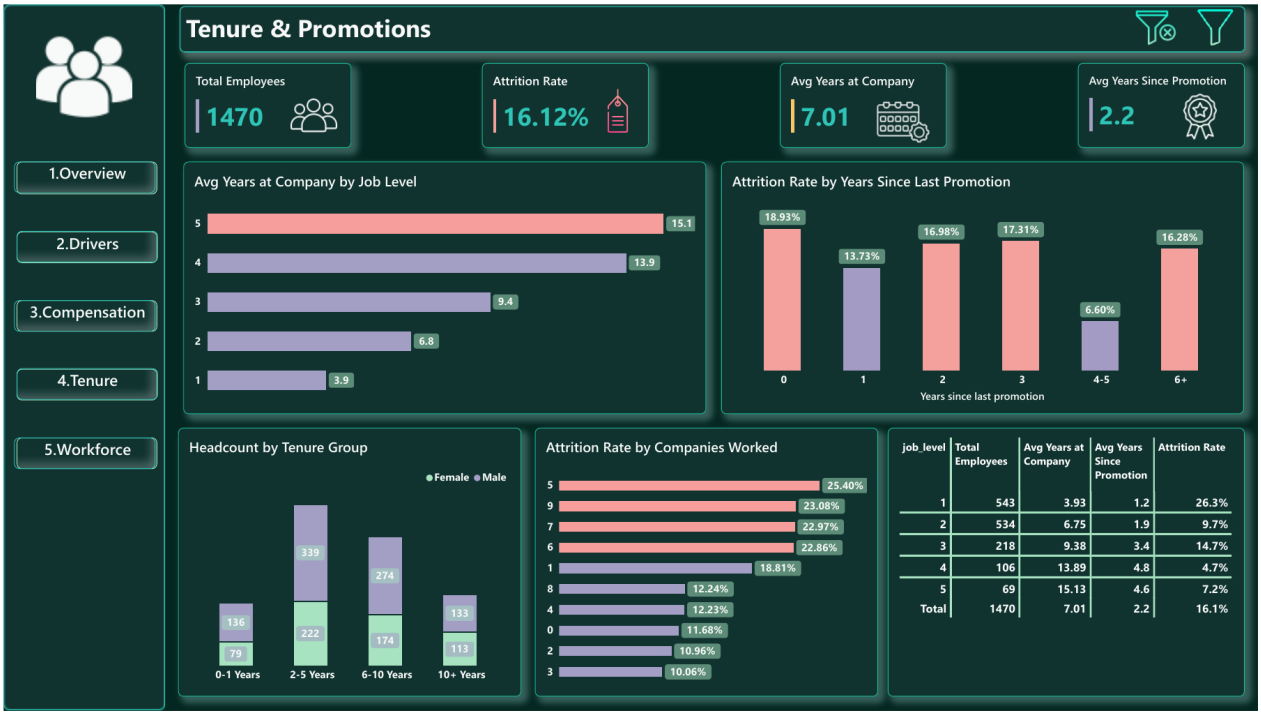


Figure 5 – Tenure & Promotions: tenure, time since promotion and companies worked.

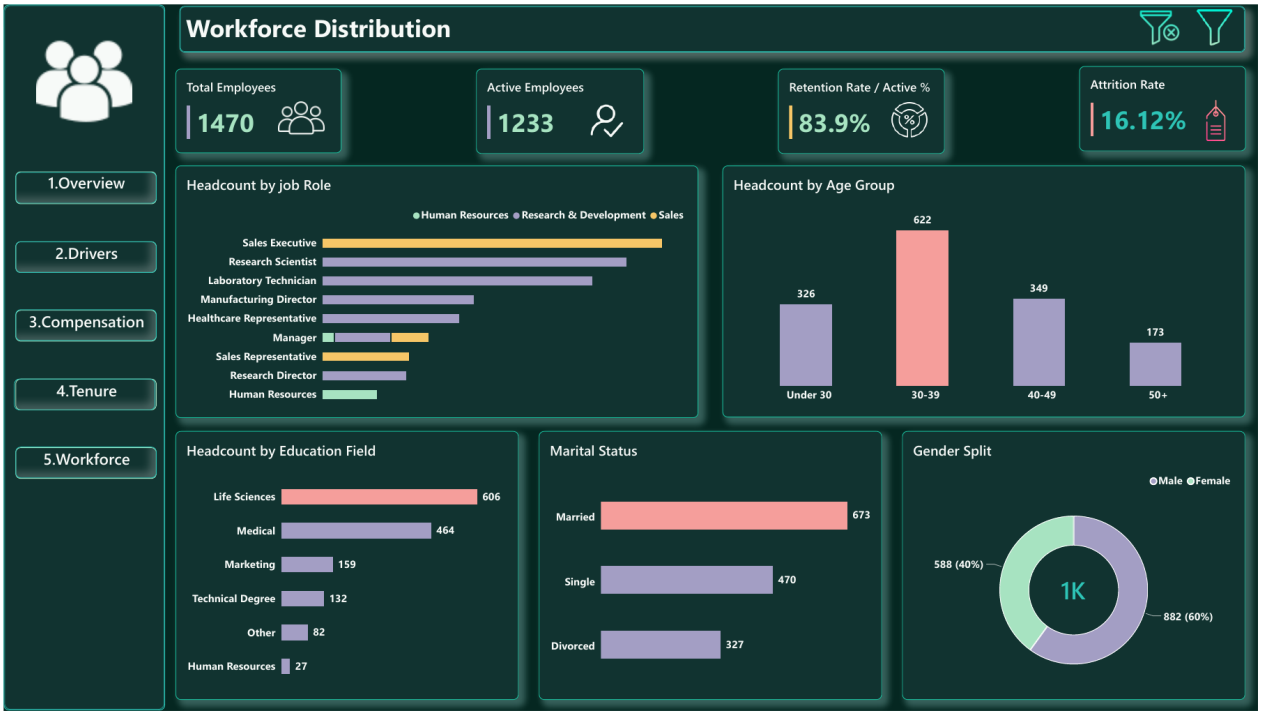


Figure 6 – Workforce Distribution: headcount by role, age, education, marital status and gender.



Figure 7 – Key Business Insights: eight findings with recommended actions.

8. Key Findings

#	Finding	Evidence
1	The first year is the riskiest	34.9% attrition for 0-1 years vs 8.1% for 10+ years (75 of 215 new hires left).
2	Overtime almost triples attrition	30.5% with overtime vs 10.4% without. Overtime workers are 28% of staff but 54% of leavers (127 of 416).
3	Sales Representatives lead exits	39.8% (33 of 83) vs 16.1% company average. Next: Laboratory Technician 23.9%. Lowest: Research Director 2.5%.
4	Sales and HR exceed the average	Sales 20.6% (92 of 446), HR 19.0%, R&D; 13.8%. R&D; is the largest department and has the lowest rate.
5	Low pay, high attrition	28.6% under $3K vs 8.9% at $10K+. Leavers averaged $4,787 vs $6,833 for stayers (about 30% lower).
6	Younger staff leave most	27.9% under 30 (91 of 326) vs 9.7% at 40-49, with a small rebound to 13.3% at 50+.
7	Highest-risk segment: single + overtime	49.6% (n = 131), about 1 in 2. Single alone: 25.5% vs Married 12.5%.
8	Frequent travel adds risk	24.9% for frequent travelers (69 of 277) vs 8.0% non-travel.

Other observations

- Job level 1 has 26.3% attrition vs 4.7% at level 4.
- Stock option level 0 has 24.4% attrition, levels 1-2 stay under 10%, but level 3 is 17.6%, so stock options are not a simple “more is better” effect.
- Low job satisfaction (level 1) shows 22.8% attrition vs 11.3% at “Very High”.

9. Recommendations

Area	Recommended action
Onboarding	Strengthen onboarding and check-ins during months 1-12.
Workload	Rebalance overtime allocation and workloads.
Sales Representatives	Review targets, pay mix and career path for this role.
Sales & HR	Pay alone does not explain Sales exits: look at workload and role.
Compensation	Benchmark entry-level salaries and review pay bands.
Early career	Offer growth paths and mentoring for staff under 30.
Highest-risk segment	Target single employees working overtime first with retention measures.
Travel & equity	Limit heavy travel loads; review stock-option eligibility.

10. Limitations

- **Snapshot data:** the dataset has no time dimension, so we cannot measure trends or how attrition changed over time.
- **Correlation, not causation:** the dashboard shows which groups leave more, not proof of why they leave.
- **Small groups:** HR (n = 63), Sales Representatives (n = 83) and the single + overtime segment (n = 131) are small; read their rates with caution.
- **Reasons for leaving are not recorded:** attrition is a Left/Stayed flag only.
- **Data origin:** if the dataset is the public IBM HR sample, it is synthetic and results should not be generalized to a real company.
- **Band definitions:** results depend on how age, income and tenure bands were defined.

TODO – team input needed: add any other limits the team found (e.g. columns removed during cleaning).

11. Conclusion

The company lost 237 of 1,470 employees (16.12%). Attrition is concentrated among **new hires, overtime workers, low earners, younger and single employees, frequent travelers and Sales Representatives**. The highest-risk segment, single employees working overtime, loses about half of its people.

The dashboard gives HR an interactive way to see these patterns and the recommendations give clear first actions: improve first-year onboarding, rebalance overtime, review entry-level pay and the Sales Representative role, and target the highest-risk segment with retention measures.